



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SAI VENKATESWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

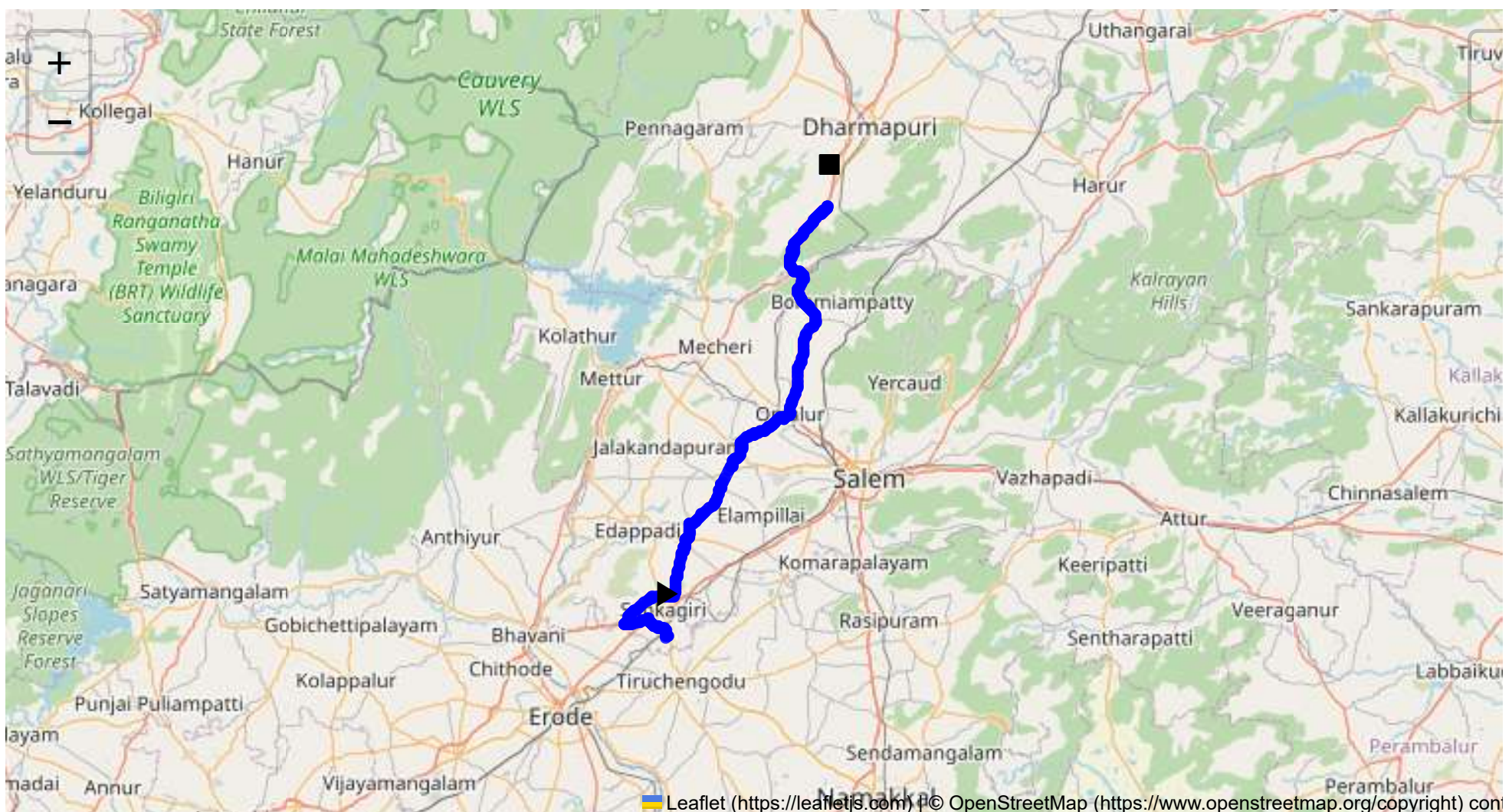
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 90.94 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours
Start: (11.4381, 77.8734)
End: (12.024528, 78.098427)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The specified route begins in Sangagiri and travels through several key locations, including Puthur, Padaiveedu, and Thoppur before reaching Palayampudur. This route covers major highways such as the Salem - Kochi Hwy and SH 86. The route spans approximately 90.94 kilometers and is a combination of state highways and local roads, with varying road conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences a tropical climate. The region can face heavy rainfall during the monsoon seasons (from June to September and October to December). Such weather may lead to reduced visibility and slippery road conditions, particularly affecting trucking operations. Dust storms are less common but could occasionally be experienced near arid locations.

3. Traffic Patterns, Peak Hours, and Congestion-Prone Areas

Traffic is generally lighter in rural areas but can get congested near towns, especially in places like Omalur and Sangagiri during peak commute hours (8 AM to 10 AM and 5 PM to 7 PM). The Bhavani Main Road and Salem - Kochi Hwy are significant thoroughfares that may experience slow-moving traffic, particularly near intersections and market areas.

4. Assessment of Road Quality and Infrastructure

Road quality varies with some stretches on the Salem - Kochi Hwy being well-maintained while smaller roads can be narrow with occasional potholes and uneven surfaces. Pay attention to road signage and speed regulations as some rural sectors may not be adequately marked.

5. Suggestions for Alternative Routes for Emergencies

In case of an emergency, one could consider diverting to national highways or major state highways where emergency services are more accessible and road conditions are typically better. For instance, highways like NH 44 or NH 79 can offer faster alternatives in case of road blockages.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Local regulations require that trucks carrying hazardous materials are appropriately marked and follow specified routes to avoid densely populated areas where possible. It is crucial to comply with speed limits and abide by road safety measures, especially in populated areas and near schools or hospitals.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been few significant incidents on these routes involving heavy vehicles, but there is always potential for accidents in high-traffic areas or due to challenging weather conditions. Review recent accident records before departure to assess current risk levels.

8. Environmental Considerations and Sensitive Areas

The route crosses agricultural areas and small communities, so there is an elevated risk of environmental impact in the case of hazardous material leakage. Ensure all materials are securely contained to minimize risk.

9. Analysis of Communication Coverage, Noting Potential Dead Zones

Communication coverage is generally reliable along major highways but can be intermittent in more remote and rural areas along the route. It is advisable to have offline maps and ensure communication devices are charged.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response times can vary. In major towns like Sangagiri and Salem, response times could be 20-30 minutes, while rural regions might experience delays extending up to an hour depending on accessibility and weather conditions.

12. Overall Summary of Risk Assessment

The route presents moderate risk primarily due to variable road quality and potential weather-related disruptions. Traffic and congestion are concerns near urban areas, while rural sections pose risks of limited amenities and services. Regular updates on weather conditions and traffic, as well as adherence to safety and regulatory guidelines, will aid in minimizing risks. Ensure all safety measures for the transport of hazardous materials are observed diligently.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45543, 77.81740	30 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	High	11.49383, 77.88442	15 KM/Hr
11	Turn	Medium	11.56083, 77.89921	30 KM/Hr
12	Turn	Medium	11.56180, 77.90019	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.57048, 77.90224	30 KM/Hr
14	Turn	Medium	11.64217, 77.95200	30 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	High	11.67858, 77.96834	15 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72559, 78.01758	30 KM/Hr
21	Turn	Medium	11.73787, 78.03528	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74120, 78.04588	30 KM/Hr
24	Turn	Medium	11.74309, 78.04725	30 KM/Hr
25	Turn	Medium	11.87913, 78.08206	30 KM/Hr
26	Turn	Medium	11.95803, 78.04964	30 KM/Hr
27	Turn	Medium	11.95854, 78.04990	30 KM/Hr
28	Turn	Medium	11.96015, 78.04945	30 KM/Hr
29	Turn	Medium	11.96370, 78.05142	30 KM/Hr
30	Turn	Medium	11.96433, 78.05113	30 KM/Hr
31	Turn	High	11.96542, 78.04938	15 KM/Hr
32	Turn	High	11.96650, 78.04927	15 KM/Hr
33	Turn	Medium	11.96804, 78.05080	30 KM/Hr
34	Blind Spot	Blind Spot	12.02456, 78.09864	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57048, 77.90224



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72559, 78.01758



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74309, 78.04725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96015, 78.04945

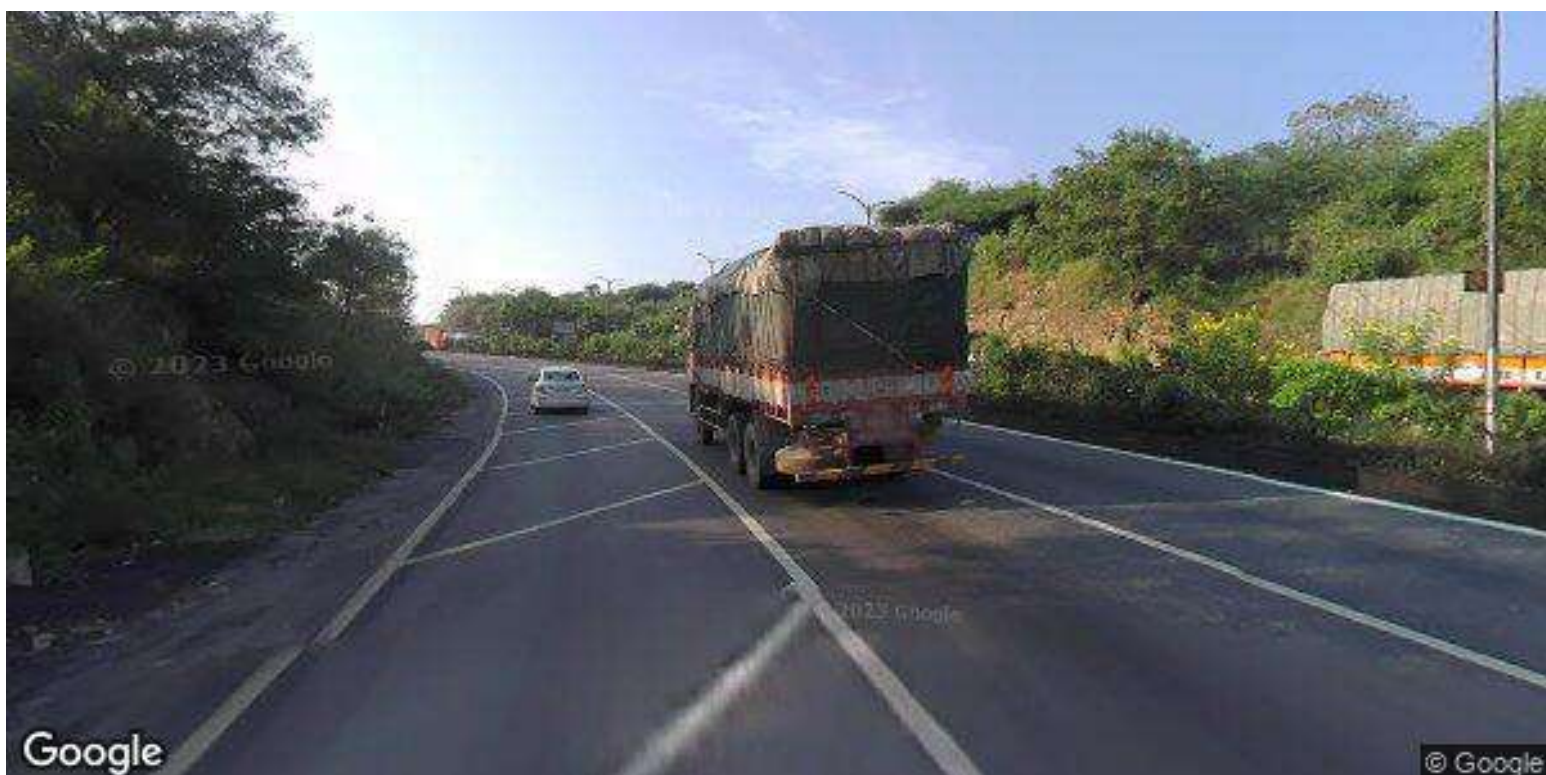


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96804, 78.05080



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.02456, 78.09864

